Skills Summary

Fraction Models: Are the Parts Equal?

Use this reference while completing your worksheet or when studying.

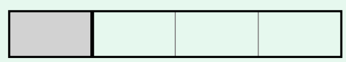
1. Check the parts first (equal-parts-check)

The rule: a fraction names *equal* parts of one whole.

Before you write any fraction, ask: **are all the parts the same size?**

- **Yes** → the model names a fraction. Bottom number = how many equal parts fill the whole; top number = how many are shaded.
- **No** → counting pieces gives the *wrong* fraction. Look for smaller equal parts inside the model and count those instead.

Example: Sam broke this strip into two pieces and says the shaded piece is one half.



Step 1. The two pieces are not the same size, so "one out of two pieces" is not a half — switch to the equal small parts the thin lines show.
Step 2. The whole strip is 4 equal small parts and the shaded piece covers 1 of them. ⇒ the shaded piece is $\frac{1}{4}$, not one half.

Try it: A strip is made of 6 equal small parts. It is broken into two pieces, and the shaded piece covers 2 of the small parts. What fraction of the strip is shaded?

check: $\frac{2}{6} = \frac{1}{3}$

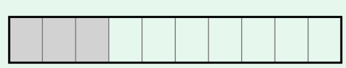
2. Name the fraction (name-fraction-from-equal-units)

How to write it:

$$\frac{\text{how many equal parts are shaded}}{\text{how many equal parts fill the whole}}$$

Count *every* equal part in the whole for the bottom number — including the ones that are not shaded.

Example: What fraction of this bar is shaded?



Step 1. All the parts are the same width, so this model does name a fraction — now just count.
Step 2. There are 10 equal parts in the whole bar, and 3 of them are shaded. ⇒ $\frac{3}{10}$ of the bar is shaded.

Try it: A bar is cut into 6 equal parts and 5 of them are shaded. What fraction of the bar is shaded?

check: $\frac{5}{6}$

3. Fix a model that has unequal parts (fix-the-model)

To show halves, thirds, fourths . . . cut the whole into that many parts *of the same size.*
If the whole is already made of equal small squares:

$$\text{squares in each part} = \text{squares in the whole} \div \text{number of parts}$$

Example: A bar of 12 equal squares must be cut into fourths. Where do the cuts go?
Step 1. Fourths means 4 parts that are all the same size, so share the squares out evenly instead of guessing where to cut.
Step 2. $12 \div 4 = 3$, so each fourth is 3 squares wide (cut after the 3rd, 6th and 9th square).
$\Rightarrow$ each fourth holds **3** squares.

Try it: A bar of 10 equal squares is cut into 2 equal halves. How many squares are in each half?

check: **5**

4. More equal parts means smaller parts

Same whole, more parts, smaller pieces. If two identical wholes are cut into equal parts, the one cut into *more* parts has the *smaller* pieces — there is only so much to share out.

Example: Two identical ribbons. One is cut into thirds, the other into fifths. Which single piece is longer?
Step 1. Both wholes are the same length, so compare the number of equal parts rather than the numbers on the bottom of the fractions.
Step 2. Thirds share the ribbon among fewer parts, so each third is longer than each fifth.
$\Rightarrow$ $\frac{1}{3} > \frac{1}{5}$

Try it: Two identical ribbons, one cut into eighths and one cut into sixths. Write > or < between the two pieces.

check: $\frac{1}{8} < \frac{1}{6}$

(!) Watch out: counting pieces is not the same as counting equal parts. A shape cut into three *unequal* pieces has no thirds in it at all.